MTH631 - Real Analysis II (Quiz No. 4)

Question # 6 of 10 (Start time: 05:57:37 PM, 07 January 2026)

If $f(x, y)$ is continuous at (x_0, y_0) and $f_{yx}(x_0, y_0)$ exists. Then,

Select the correct option

○	$f_{xx}(x_0, y_0) = f_{yy}(x_0, y_0)$
●	$f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0)$
○	None of these
○	$f_x(x_0, y_0) = f_y(x_0, y_0)$

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Question # 4 of 10 (Start time: 05:57:18 PM, 07 January 2026)

In $\mathbb{R}^2$, $\lim_{(x,y)\to(2,2)} \frac{\sin(x-y)}{\sqrt{x-y}} = ----.$

Select the correct option

○	1
○	$\sqrt{2}$
●	0
○	∞

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Question # 3 of 10 (Start time: 05:57:09 PM, 07 January 2026)

If $f(x, y, z) = \cos\left(\frac{1}{x^2+2y^2+z^2}\right)$ then $\lim_{|X|\to\infty} f(X) =$ _____.

Select the correct option

◉	1
○	infinite
○	0
○	-1

Question # 5 of 10 (Start time: 05:57:27 PM, 07 January 2026) Total Marks: 1

Let f be defined and continuous on a region S in $\mathbb{R}^n$. Suppose that X_1 and X_2 are in S and $f(X_1) < Y < f(X_2)$.

Then $f(X) = Y$ - - - - - - - X in S.

Select the correct option

Reload Math Equations

- for some
- for all

Click to Save Answer & Move to Next Question

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Question # 2 of 10 (Start time: 05:56:58 PM, 07 January 2026)

Let the function $f(s) = \begin{cases} \frac{\sin s}{s}, s \neq 0 \\ 1, s = 0 \end{cases}$ is continuous over domain _____.

Select the correct option

- (b) $(-\infty, \infty)$
- Both a & b
- (c) $(0, \infty)$
- (a) $\mathbb{R}$

Question # 8 of 10 (Start time: 05:58:09 PM, 07 January 2026)

In $\mathbb{R}^n$, if the first order partial derivative of a function $f(X)$ exits at X_0, then it is essentially continuous at X_0.

Select the correct option

- (●) True
- ( ) False

Question # 9 of 10 (Start time: 05:58:19 PM, 07 January 2026) Total Marks: 1

In $\mathbb{R}^n$, the function $f(X)$ is continuous at X_0, if $\lim_{X \to X_0} f(X) = f(X_0)$, then - - - - - - .

Select the correct option

Reload Math Equations

- ( ) neither X_0 is in the domian D_f nor the limit point of D_f
- ( ) X_0 is in the domian D_f but not a limit point of D_f
- ( ) X_0 is not in the domian D_f but a limit point of D_f
- (●) X_0 is in the domian D_f and limit point of D_f

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 05:58:40 PM, 07 January 2026) Total Marks: 1

Let u and v be functions of two variables with continuous second - order partial derivatives in a region S.
Suppose that $u_x = v_y$ and $u_y = -v_x$ in S. Then, $u_{xx} + u_{yy} - - - -$.

Select the correct option

Reload Math Equations

- >0 but <1
- =1
- >0
- =0 (selected)

Click to Save Answer & Move to Next Question

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Question # 1 of 10 (Start time: 05:56:46 PM, 07 January 2026)

The domain of the real valued function $f : \mathbb{R}^n \to \mathbb{R}$ defined and given by;
$f(X) = \left(1 - x_1^2 - x_2^2 - \cdots - x_2^n\right)^{-1}$ is - - - - - .

Select the correct option

- $\{X \mid |X| \neq 1\}$ (selected)
- $\{X \mid |X| = 1\}$

Question # 7 of 10 (Start time: 05:57:55 PM, 07 January 2026)

$$\lim_{(x,y)\to(a,b)} \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}} = 1, \text{ if} - - - - - - .$$

Select the correct option

○	$(a^2+2b^2) > 1$
◉	$(a^2+2b^2) = 1$
○	$(a^2+2b^2) < 1$
○	$(a^2+2b^2) \gtreqless 1$